GRADE 11 / Physical Sciences / English

Forces and Newton’s second law

01 Understand

Learning goals: Draw a free-body diagram • Calculate a net horizontal force • Apply Fnet = ma with a sign convention

A free-body diagram isolates one object and shows forces acting on it, not forces it exerts on other objects.

Choose a positive direction. Resolve and add forces along that axis before applying Fnet = ma.

On a horizontal surface with no vertical acceleration, normal force balances weight if there are no other vertical forces. Friction acts against relative motion or its tendency.

Free-body inventory: 5 kg crate pulled horizontally

Force	Direction	Magnitude
Pull	right	20 N
Friction	left	5 N
Weight	down	49 N
Normal	up	49 N

02 Follow the reasoning

Find the horizontal acceleration of the crate. Use g = 9.8 m·s^{-2}.

1. Take right as positive. Vertical resultant = 49 − 49 = 0 N.

2. Horizontal resultant = 20 − 5 = 15 N right.

3. a = Fnet/m = 15/5 = 3 m·s^{-2} right.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. A 4 kg trolley has a net force of 12 N left. Find acceleration.

2. What resultant force keeps an object at constant velocity?

3. Are weight and normal force a Newton's third-law pair? Explain.

04 Check and correct

1. 3 $m \cdot s^{-2}$ left.

2. Zero resultant force; individual forces may still act.

3. No. They act on the same object. A third-law pair acts on two different objects.

Common mistake to avoid

Never use the applied force alone in Fnet = ma when friction or other forces also act.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Mechanics: Newton's laws and application

DBE CAPS, Grades 10–12: Physical Sciences. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID phsc-11.